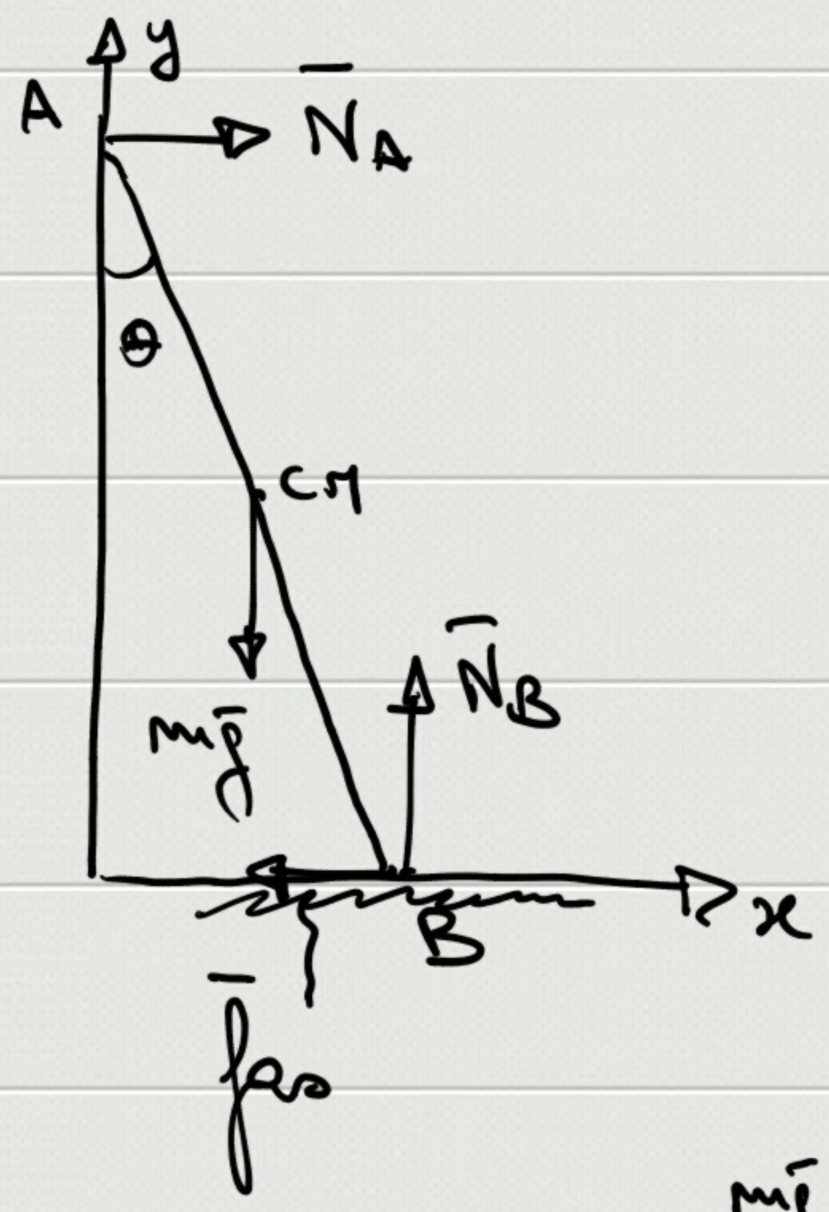
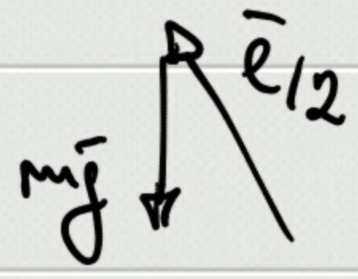
$$\bar{R}^{E} = m\bar{a}_{CM} \qquad \bar{M}_{O}^{E} = \bar{I}_{z}\,\bar{\alpha}$$

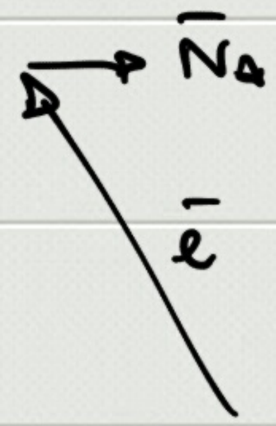
$$v_O = \omega_O = 0 \qquad \bar{R}^{E} = \bar{M}_{O}^{E} = 0 \Rightarrow \underline{\underline{\text{statica}}}$$



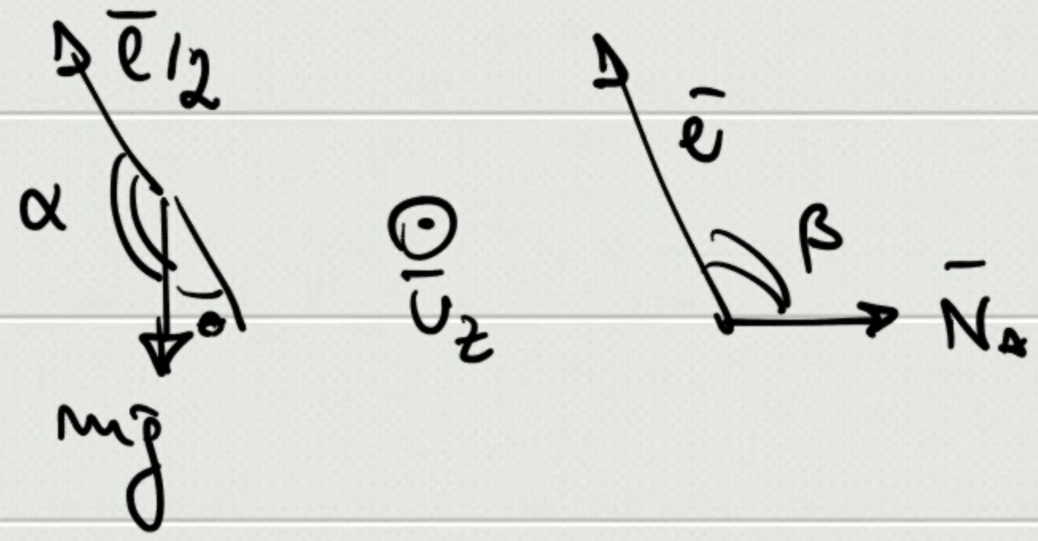
$AB = \ell$, m

$$\bar{R}^{E} = 0 \begin{cases} x : \underbrace{N_A = f_{as}} \\ y : N_B = mg \end{cases}$$





$$\bar{M}_{B}^{E} = 0 \Rightarrow \frac{\bar{\ell}}{2} \times m\bar{g} + \bar{\ell} \times \bar{N}_A = 0$$

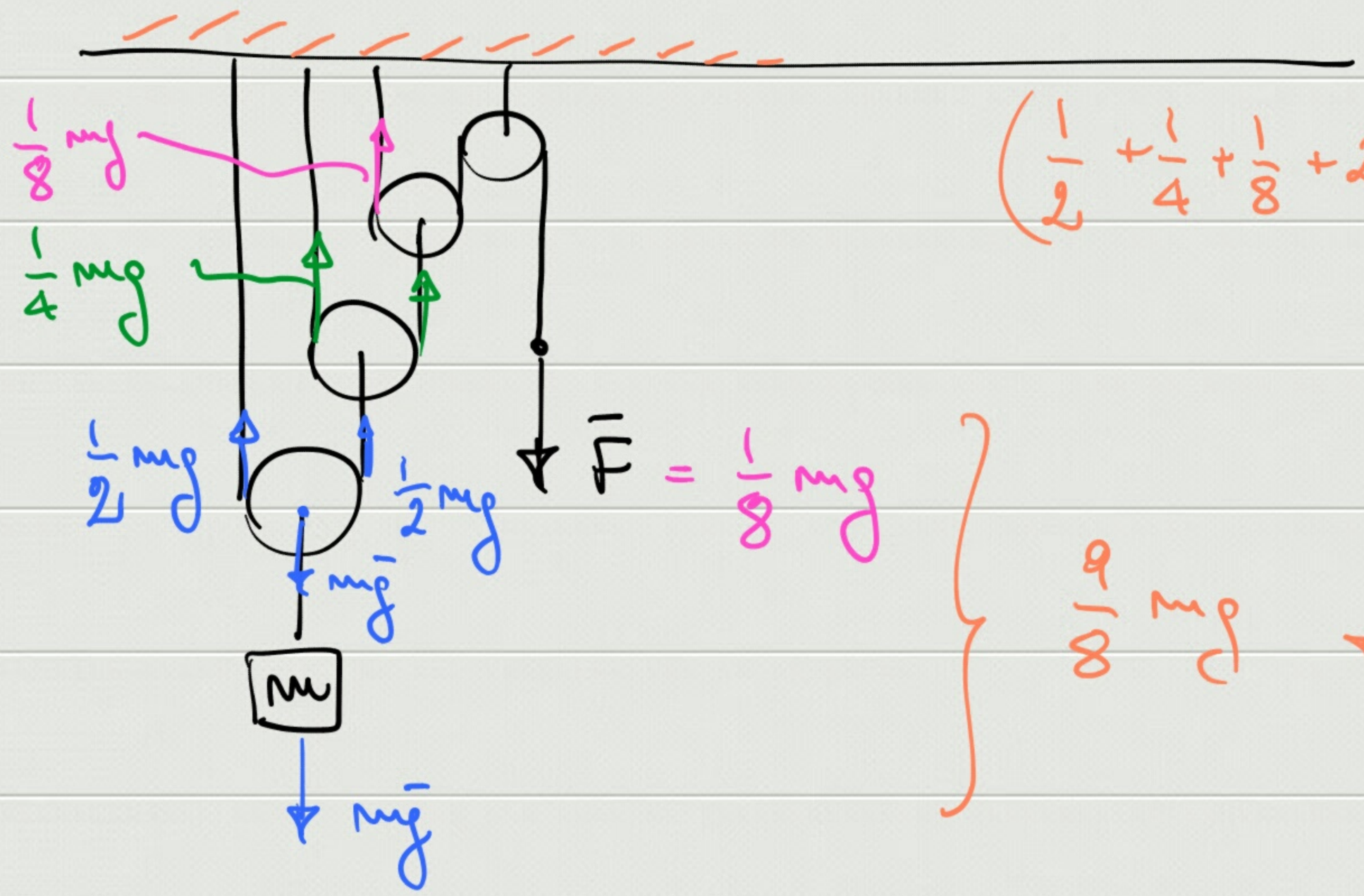


$$\Rightarrow \frac{\cancel{\ell}}{2} mg \sin(\pi - \theta)\,\bar{u}_z + \cancel{\ell} N_A \sin\left(\frac{\pi}{2} + \theta\right)(-\bar{u}_z) = 0$$

$$\Rightarrow \frac{1}{2} mg \sin\theta - N_A \cos\theta = 0 \Rightarrow N_A = \frac{1}{2} mg\, tg\,\theta$$

$$\left.\begin{aligned} N_A &= \frac{1}{2} mg \operatorname{tg}\theta \\ &\overset{!}{=} f_{as} \leq \mu_s N_B = \mu_s mg \end{aligned}\right\} \Longrightarrow$$

$$\Rightarrow \frac{1}{2} \cancel{m}g \operatorname{tg}\theta \leq \mu_s \cancel{m}g \Rightarrow \boxed{\mu_s \geq \frac{1}{2} \operatorname{tg}\theta}$$



$$\left(\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + 2 \cdot \frac{1}{8}\right) mg = \frac{9}{8} mg$$

$$\frac{9}{8} mg \downarrow$$